MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC ASSESSMENT – 08** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

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Q1. $(\sec A + \tan A)(1 - \sin A)$ is equal to: [1]

(A) $\sec A$ (B) $\sin A$ (C) $\csc A$ (D) $\cos A$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $\cos A$

Justification: $\left(\frac{1}{\cos A} + \frac{\sin A}{\cos A}\right)(1 - \sin A) = \frac{1 - \sin^2 A}{\cos A} = \frac{\cos^2 A}{\cos A} = \cos A.$

Q2. $\frac{1 + \tan^2 A}{1 + \cot^2 A}$ is equal to: [1]

(A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $\tan^2 A$

Justification: $\frac{1 + \tan^2 A}{1 + \cot^2 A} = \frac{\sec^2 A}{\csc^2 A} = \frac{1/\cos^2 A}{1/\sin^2 A} = \frac{\sin^2 A}{\cos^2 A} = \tan^2 A.$

Q3. If $\cos\theta + \cos^2\theta = 1$, then the value of $\sin^2\theta + \sin^4\theta$ is: [1]

(A) -1 (B) 0 (C) 1 (D) 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 1

Justification: Since $\cos\theta = 1 - \cos^2\theta = \sin^2\theta$, we have $\sin^2\theta + \sin^4\theta = \cos\theta + \cos^2\theta = 1$.

Q4. If $\sec\theta + \tan\theta = x$, then the value of $\sec\theta - \tan\theta$ is: [1]

(A) x^2 (B) $\frac{1}{x}$ (C) $-x$ (D) $1 - x$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $\frac{1}{x}$

Justification: From $\sec^2\theta - \tan^2\theta = 1$, $(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1 \implies \sec\theta - \tan\theta = \frac{1}{x}$.

Q5. Using the previous result, if $\sec\theta + \tan\theta = x$, then $\sec\theta$ is equal to: [1]

(A) $\frac{x^2+1}{2x}$ (B) $\frac{x^2-1}{2x}$

(C) $\frac{x^2+1}{x}$ (D) $\frac{2x}{x^2+1}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{x^2+1}{2x}$

Justification: Adding $\sec\theta + \tan\theta = x$ and $\sec\theta - \tan\theta = \frac{1}{x}$ gives $2\sec\theta = x + \frac{1}{x} = \frac{x^2+1}{x} \implies \sec\theta = \frac{x^2+1}{2x}$.

Q6. **Assertion (A):** The value of $\frac{1-\cos^2\theta}{\sin^2\theta}$ is equal to 1 for all acute angles θ.
Reason (R): $\sin^2\theta + \cos^2\theta = 1 \implies 1 - \cos^2\theta = \sin^2\theta$, and for acute angles θ, $\sin\theta \neq 0$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

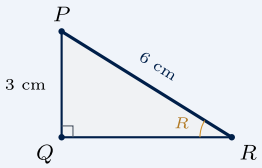
Justification: For $0° < \theta < 90°$, $\sin\theta \neq 0$, so $\frac{1-\cos^2\theta}{\sin^2\theta} = \frac{\sin^2\theta}{\sin^2\theta} = 1$. Both are true and (R) explains (A).

Q7. In $\triangle PQR$, right-angled at Q, $PQ = 3$ cm and $PR = 6$ cm. Determine $\angle QPR$ and $\angle PRQ$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Find $\angle PRQ$ [1 Mark]

$$\sin(\angle PRQ) = \frac{PQ}{PR} = \frac{3}{6} = \frac{1}{2} = \sin 30^\circ \implies \angle PRQ = 30^\circ$$



Step 2: Find $\angle QPR$ [1 Mark]

By angle sum property of $\triangle PQR$, $\angle QPR = 180^\circ - 90^\circ - 30^\circ = 60^\circ$.

$$\boxed{\angle QPR = \mathbf{60^\circ}, \qquad \angle PRQ = \mathbf{30^\circ}}$$

Q8. If $\cos\theta = \frac{1}{2}$, find the value of $\frac{2\sec\theta}{1+\tan^2\theta}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Apply Identity $1 + \tan^2\theta = \sec^2\theta$ [1 Mark]

$$\frac{2\sec\theta}{1+\tan^2\theta} = \frac{2\sec\theta}{\sec^2\theta} = \frac{2}{\sec\theta} = 2\cos\theta$$

Step 2: Substitute $\cos\theta = \frac{1}{2}$ [1 Mark]

$$= 2 \times \frac{1}{2} = \boxed{\mathbf{1}}$$

Q9. Prove that: $\dfrac{\cos^3\theta + \sin^3\theta}{\cos\theta + \sin\theta} + \dfrac{\cos^3\theta - \sin^3\theta}{\cos\theta - \sin\theta} = 2.$ [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Factor Cubes $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$ [1.5 Marks]

$$\text{LHS} = \frac{(\cos\theta + \sin\theta)(\cos^2\theta - \cos\theta\sin\theta + \sin^2\theta)}{\cos\theta + \sin\theta} + \frac{(\cos\theta - \sin\theta)(\cos^2\theta + \cos\theta\sin\theta + \sin^2\theta)}{\cos\theta - \sin\theta}$$

Step 2: Cancel Common Factors & Simplify [1.5 Marks]

$$\text{LHS} = (\cos^2\theta + \sin^2\theta - \cos\theta\sin\theta) + (\cos^2\theta + \sin^2\theta + \cos\theta\sin\theta) = 1 + 1 = \boxed{\mathbf{2} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q10. Prove that: $\dfrac{\sin A}{1-\cos A} = \csc A + \cot A$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Multiply Numerator & Denominator by $(1+\cos A)$ [1.5 Marks]

$$\text{LHS} = \frac{\sin A}{1-\cos A} \times \frac{1+\cos A}{1+\cos A} = \frac{\sin A(1+\cos A)}{1-\cos^2 A} = \frac{\sin A(1+\cos A)}{\sin^2 A}$$

Step 2: Simplify to RHS [1.5 Marks]

$$\text{LHS} = \frac{1+\cos A}{\sin A} = \frac{1}{\sin A} + \frac{\cos A}{\sin A} = \boxed{\csc \mathbf{A} + \cot \mathbf{A} = \text{RHS} \quad \text{[Hence Proved]}}$$

SECTION D: Long Answer / HOTS

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. Prove that: $\dfrac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$. **[5]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Divide Numerator & Denominator by $\sin A$ [1.5 Marks]

$$\text{LHS} = \frac{\frac{\cos A}{\sin A} - \frac{\sin A}{\sin A} + \frac{1}{\sin A}}{\frac{\cos A}{\sin A} + \frac{\sin A}{\sin A} - \frac{1}{\sin A}} = \frac{\cot A + \csc A - 1}{\cot A - \csc A + 1}$$

Step 2: Replace 1 **in Numerator with** $\csc^2 A - \cot^2 A$ [2 Marks]

$$\text{Num} = (\cot A + \csc A) - (\csc^2 A - \cot^2 A) = (\cot A + \csc A) - (\csc A - \cot A)(\csc A + \cot A)$$

$$\text{Num} = (\csc A + \cot A)(1 - \csc A + \cot A)$$

Step 3: Cancel Common Factor $(\cot A - \csc A + 1)$ [1.5 Marks]

$$\text{LHS} = \frac{(\csc A + \cot A)(1 - \csc A + \cot A)}{1 - \csc A + \cot A} = \boxed{\csc \mathbf{A} + \cot \mathbf{A} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q12. If $x = a\cos^3\theta$ and $y = b\sin^3\theta$, prove that $\left(\dfrac{x}{a}\right)^{2/3} + \left(\dfrac{y}{b}\right)^{2/3} = 1$. **[5]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Isolate $\cos^3\theta$ **and** $\sin^3\theta$ [1.5 Marks]

$$\frac{x}{a} = \cos^3\theta \qquad \text{and} \qquad \frac{y}{b} = \sin^3\theta$$

Step 2: Raise Both to Power $\frac{2}{3}$ [2 Marks]

$$\left(\frac{x}{a}\right)^{2/3} = \left(\cos^3\theta\right)^{2/3} = \cos^2\theta, \qquad \left(\frac{y}{b}\right)^{2/3} = \left(\sin^3\theta\right)^{2/3} = \sin^2\theta$$

Step 3: Add & Apply $\cos^2\theta + \sin^2\theta = 1$ [1.5 Marks]

$$\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = \cos^2\theta + \sin^2\theta = \boxed{\mathbf{1} \quad \text{[Hence Proved]}}$$

Q13. Mountain Cable Car / Aerial Tramway Span

A passenger ropeway connects a valley base station V to an alpine mountain peak station M. The cable runs straight between the two stations along an incline angle of $\theta = 30°$ to the horizontal. The vertical elevation rise between the two stations is $h = 500$ m. **Based on the above geometric model, answer the following questions:**

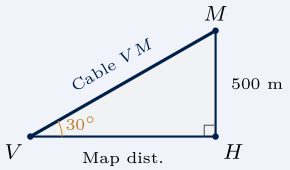
(i) Find the total direct track length VM of the ropeway cable. [1]

(ii) Find the horizontal map distance between the base station and mountain station. [1]

(iii) If a cable car travels at a constant speed of 5 m/s, calculate the travel time in seconds and minutes from the valley station to the summit. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Direct Track Length VM [1 Mark]



$$\sin 30° = \frac{MH}{VM} \implies \frac{1}{2} = \frac{500}{VM} \implies \boxed{VM = \mathbf{1000}\text{ m}}$$

Sub-part (ii): Horizontal Map Distance VH [1 Mark]

$$\tan 30° = \frac{500}{VH} \implies \frac{1}{\sqrt{3}} = \frac{500}{VH}$$

$$\implies \boxed{VH = \mathbf{500\sqrt{3}}\text{ m} \approx 866.03\text{ m}}$$

Sub-part (iii): Travel Time at 5 m/s [2 Marks]

$$\text{Time} = \frac{VM}{5} = \frac{1000}{5} = \mathbf{200}\text{ s} = \frac{200}{60}\text{ min}$$

$$\implies \boxed{\text{Time} = \mathbf{200}\text{ s}\ \left(\mathbf{3}\text{ min }\mathbf{20}\text{ s or }\mathbf{3\frac{1}{3}}\text{ min}\right)}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★